Table of 15 key Windows Security Event IDs with meaning and when to use each.

Important Windows Security Event IDs

Windows Security Event IDs are numerical identifiers that describe specific security-related activities.

SOC analysts use them to investigate:

- Failed and successful authentication.
- Suspicious account activity.
- Privilege escalation.
- Account creation.
- Process execution.
- Lateral movement.
- Credential abuse.
- Log tampering.

The following Event IDs are particularly important.

1. Windows Security Event ID reference table

Event ID	Meaning	When should a SOC analyst use it?
4624	Successful account logon	Identify who logged in, when, from where, and using which logon type.
4625	Failed account logon	Investigate brute-force attacks, password spraying, and repeated authentication failures.
4634	Account logged off	Identify when a logon session ended.
4648	Logon attempted using explicit credentials	Investigate use of alternate credentials, especially suspicious runas or remote access activity.
4672	Special privileges assigned to a new logon	Identify logons receiving sensitive privileges, such as administrative privileges.
4688	New process created	Investigate suspicious process execution, command lines, and parent-child relationships.
4697	A service was installed in the system	Investigate unauthorized service installation and potential persistence.
4720	User account created	Identify unauthorized account creation.
4722	User account enabled	Investigate unexpected enabling of disabled accounts.
4724	An attempt was made to reset an account's password	Investigate unauthorized password resets.
4732	A member was added to a security-enabled local group	Investigate suspicious additions to the local Administrators group or other privileged groups.

4768	Kerberos authentication ticket (TGT) requested	Investigate Kerberos authentication activity and possible AS-REP roasting indicators.
4769	Kerberos service ticket requested	Investigate service-ticket activity, unusual access, and possible Kerberoasting.
4776	Computer attempted to validate account credentials using NTLM	Investigate NTLM authentication failures, brute-force attempts, and suspicious NTLM usage.
1102	Security audit log cleared	Investigate possible log tampering and attempts to hide malicious activity.

2. Detailed explanation of important Event IDs

i. Event ID 4624 - Successful Logon

Successful authentication

This event is generated when an account successfully logs on.

Important fields:

- Account Name.
- Account Domain.
- Logon Type.
- Source Network Address.
- Workstation Name.
- Authentication Package.
- Logon ID.

Example:

Event ID: 4624
Account Name: john
Logon Type: 10
Source Network Address: 192.168.1.50
Authentication Package: Negotiate

Interpretation:

The user john successfully logged in using Remote Desktop from 192.168.1.50.

SOC investigation:

- Is this user's normal workstation?
- Is the login time normal?
- Is the source IP authorized?
- Does the account usually use Remote Desktop?
- Was the account involved in suspicious activity afterward?

ii. Event ID 4625 - Failed Logon

Authentication failure

Generated when an account logon attempt fails.

Example:

Event ID: 4625
Account Name: administrator
Logon Type: 3
Source Network Address: 192.168.1.100
Failure Reason: Unknown user name or bad password

Potential attack scenarios:

- Brute-force attack.
- Password spraying.
- Credential stuffing.
- Unauthorized remote access.

SOC investigation:
Check whether multiple failures originate from one IP or whether one account receives failures from multiple IPs.
A high volume of failures followed by a successful login may be particularly important.

iii. Event ID 4634 - Logoff

Generated when a logon session is terminated.

Example:

Event ID: 4634
Account Name: john
Logon ID: 0x123456

SOC investigation:
Correlate this with Event ID 4624 using the Logon ID to understand session duration.
Note that 4634 is a session logoff event; it does not necessarily tell you why the session ended.

iv. Event ID 4648 - Explicit Credentials Used

Generated when a process attempts to log on using explicitly supplied credentials.

Example:

runas /user:Administrator cmd.exe

Why important?
An attacker may use another user's credentials to access systems or execute applications.

SOC investigation:

- Which account supplied the credentials?

- Which account was targeted?
- Which process performed the action?
- What was the destination system?
- Was the action authorized?

This event can also be generated during legitimate administrative activity.

v. Event ID 4672 - Special Privileges Assigned

Generated when special privileges are assigned to a new logon.

Examples of sensitive privileges include:

- SeDebugPrivilege.
- SeBackupPrivilege.
- SeRestorePrivilege.
- SeTakeOwnershipPrivilege.

Example:
An administrator logs in and receives sensitive system privileges.

SOC investigation:
Check whether the account is expected to have those privileges.
Do not treat every 4672 event as malicious. Normal SYSTEM and administrative logons can generate it.

vi. Event ID 4688 - Process Creation

Important for endpoint investigations
Generated when a new process starts, provided the relevant auditing is enabled.

Example:

Event ID: 4688

New Process:

powershell.exe

Creator Process:

winword.exe

Process Command Line:

powershell.exe -EncodedCommand ...

This is suspicious because Microsoft Word normally should not need to launch an encoded PowerShell command.

SOC investigation:

- Examine the process name.

- Identify the parent process.
- Inspect the command line.
- Identify the user.
- Check the executable path.
- Correlate with network connections.
- Search for related file creation.

Important: Command-line visibility requires the appropriate process command-line auditing policy. The event may contain incomplete information if it is not enabled.

vii. Event ID 4697 - Service Installed

Generated when a service is installed in the system, provided the relevant auditing is configured.

Example:

Service Name: WindowsUpdateHelper

Service File Name:

C:\Users\Public\helper.exe

Suspicious because the executable is located in a publicly writable directory and the service name resembles a legitimate Windows component.

SOC investigation:

- Identify the service creator.
- Inspect the executable path.
- Check the service account.
- Determine whether the installation was approved.
- Correlate with Event ID 4688.

viii. Event ID 4720 - User Account Created

Example:

Event ID: 4720
New Account Name: support_backup
Subject Account: Administrator

SOC investigation:

Determine:

- Who created the account?
- When was it created?
- Was it authorized?
- Was it added to a privileged group?
- Was the account subsequently used?

A suspicious account created outside normal IT processes may indicate persistence.

ix. Event ID 4722 - User Account Enabled

Example:

Event ID: 4722
Account Name: old_admin

An attacker may enable an old disabled account to regain access.

Investigate the subject account responsible for enabling it and the account's subsequent authentication activity.

x. Event ID 4724 - Password Reset Attempt

Generated when an attempt is made to reset an account password.
This differs from an ordinary password-change event.

Example:
An administrator resets another user's password.

SOC investigation:

- Who performed the reset?
- Which account was targeted?
- Was the action approved?
- Did the target account subsequently log in from an unusual location?

xi. Event ID 4732 - Member Added to Local Security Group

Example:

Event ID: 4732
Group Name: Administrators
Member Name: john

This may indicate privilege escalation.

SOC investigation:

- Who added the member?
- Which account was added?
- Was it a local or domain-related administrative action?
- Was the change approved?
- Did the new administrator account perform suspicious activities afterward?

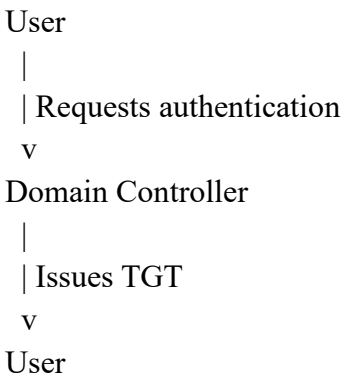
Remember: Event ID 4732 specifically concerns a security-enabled local group. Domain security group membership changes have different relevant Event IDs.

xii. Event ID 4768 - Kerberos TGT Requested

Kerberos is an authentication protocol commonly used in Active Directory environments.

A TGT (Ticket Granting Ticket) allows a user to request service tickets without repeatedly submitting their password.

Simplified authentication:



SOC investigation:

- Identify the requesting account.
- Examine the source IP.
- Look for unusual authentication patterns.
- Investigate encryption types and failure codes.
- Check for AS-REP roasting indicators.

AS-REP roasting: An attacker may target accounts configured without Kerberos pre-authentication to obtain material that can be attacked offline.

xiii. Event ID 4769 - Kerberos Service Ticket Requested

Generated when a Kerberos service ticket is requested.

Example:

A user accesses a file server using Kerberos.

The client requests a ticket for the relevant service principal name (SPN).

SOC investigation:

- Look for unusual numbers of service ticket requests.
- Identify accounts requesting tickets for many services.
- Examine encryption types.
- Investigate unusual RC4 ticket activity.
- Check for possible Kerberoasting.

Kerberoasting: An attacker requests service tickets for accounts with SPNs and attempts to crack the ticket material offline.

An unusual encryption type alone does not prove an attack; legacy systems and configuration differences can explain it.

xiv. Event ID 4776 - NTLM Credential Validation

Generated when a computer attempts to validate account credentials using NTLM.

Example:

Event ID: 4776
Authentication Package: NTLM
Account Name: administrator
Source Workstation: PC-001

SOC investigation:

- Identify repeated failed authentication attempts.
- Identify unusual source workstations.
- Investigate NTLM usage where Kerberos is expected.
- Look for potential pass-the-hash activity.

Note that this event is generally recorded on the computer authoritative for the credential validation, often a domain controller for domain accounts.

xv. Event ID 1102 - Security Audit Log Cleared

High-priority investigation
This event indicates that the Windows Security audit log was cleared.

Example:

Event ID: 1102
Subject Account: administrator

Why important?
Attackers may attempt to remove evidence of their activity.

SOC investigation:

1. Identify the account that cleared the log.
2. Determine whether it was an authorized administrator.
3. Check surrounding events.
4. Investigate suspicious logons.
5. Review other endpoint and centralized SIEM logs.

A cleared Security log is not automatically proof of an attack, but it should be investigated.

3. Event ID investigation cheat sheet

Investigation objective	Event IDs to prioritize
Brute-force attack	4625, 4776
Successful suspicious login	4624
Remote Desktop activity	4624, 4634
Alternate credential usage	4648
Privilege escalation	4672, 4732
Process execution	4688
Persistence through services	4697
Unauthorized account creation	4720
Account reactivation	4722
Password reset abuse	4724
Kerberoasting	4769
AS-REP roasting	4768
Log tampering	1102